

PROPERTIES OF A GENERALISED INACCURACY MEASURE

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Summary: In the present work, we consider a generalised version of the inaccuracy measure considered in Kerridge (1961). The behaviour of this measure in the truncated situation is examined. Further, we characterise some distributions using the functional form of the measure under the assumption that the underlying distributions follow the proportional hazard model.

1. Introduction

There has been considerable interest in recent years to study the truncated versions of various information measures and their applications in different fields such as characterisation of probability distributions, statistical inference,

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communication theory, stochastic ordering, reliability and survival analysis. Of these the Shannon's entropy of a non-negative random variable with distribution function $F(x)$ and density functions $f(x)$,

$$H(F) = - \int_0^{\infty} f(x) \log f(x) dx \quad (1.1)$$

and the Kullback-Leibler directed divergence of two non-negative random variables X and Y with respective distribution functions $F(x)$ and $G(y)$ and densities $f(x)$ and $g(y)$,

$$D(F, G) = \int_0^{\infty} f(x) \log \left(\frac{f(x)}{g(x)} \right) dx \quad (1.2)$$

have received most attention. The left truncated version of (1.1) and (1.2), obtained by replacing $f(x)$ by $f(x)/\bar{F}(t)$ and $g(x)$ by $g(x)/\bar{G}(t)$ for x in (t, ∞) were studied in Ebrahimi (1996); Ebrahimi and Kirmani (1996); Nair and Rajesh (1998); Belzunce et al. (2004) and their references. Later the results in the above works were extended to the truncated versions of the generalisations of (1.1) and (1.2) defined as

$$H_r(F) = \frac{1}{r} \int_0^{\infty} f(x) (1 - f^r(x)) dx, \quad r > -1 \quad (1.3)$$

and

$$D_r(F, G) = \frac{1}{r} \int_0^e f(x) (f^r(x) - g^r(x)) dx. \quad (1.4)$$

For the properties of these extended measures we refer to Belzunce et al. (2004); Nanda and Paul (2006) and Nair and Gupta (2007).

Kerridge (1961) introduced a measure of inaccuracy by defining it as

$$I(F, G) = - \int_0^{\infty} f(x) \log g(x) dx. \quad (1.5)$$

He attributed the inaccuracy in experimental results to (i) the uncertainty in the observations and (ii) misclassification of the model as G when it is in fact F . One can see that

$$I(F, G) = H(F) + D(F, G) \quad (1.6)$$

so that in the Kerridge inaccuracy measure the errors (i) and (ii) are accommodated by $H(F)$ and $D(F, G)$ respectively. In view of the decomposition in (1.6) and the generalisations of the respective components in (1.6) by (1.3) and (1.4), a natural question is whether there exist an extended version of the inaccuracy measure that combines (1.3) and (1.4). This motivates the definition of inaccuracy of order r and the discussion of the properties of the new measure carried out in the present paper. We further note that the generalised inaccuracy measure imparts more flexibility than (1.5) in the following sense. In the definition (1.5) all events that constitute it are assigned equal weights. In the generalised measure, the experimenter can by a judicious choice of r , give higher probabilities to events that are more sensitive to I or vice versa. Obviously when $r \rightarrow 0$, the case of equal probabilities as in (1.5) will be restored.

2. Generalized measure of inaccuracy

We define an extension of the Kerridge (1961) measure of inaccuracy as

$$I_r(F, G) = \frac{1}{r} \int_0^{\infty} f(x) [1 - g^r(x)] dx, \quad r > -1. \quad (2.1)$$

Some properties of this measure are

1. (i) As $r \rightarrow 0$, $I_r(F, G) = I(F, G)$

(ii)

$$I_r(F, G) = H_r(F) + D_r(F, G) \quad (2.2)$$

so that the new inaccuracy measure is the sum of the r -th order Shannon's entropy and the directed divergence measure. This is consistent with the decomposition and interpretation of the Kerridge measure.

(iii) $I_r(F, G)$ is minimum when $f(x) = g(x)$. In this case

$I_r(F, G) = H_r(F)$, the uncertainty in the observations alone.

(iv) $I_r(F, G) = 0$ implies $E_X(g^r(X)) = 1$ which follows from (2.1).

The truncated version of (2.1) when $X > t$, becomes

$$I_r(F, G, t) = \frac{1}{r} \int_t^\infty \frac{f(x)}{F(t)} \left[1 - \left(\frac{g(x)}{G(t)} \right)^r \right] dx. \quad (2.3)$$

For convenience we write $I_r(F, G, t) = I_r(t)$. Differentiating (2.3) with respect to t ,

$$1 - rI_r(t) = \frac{h_1(t)h_2^r(t) - rI_r'(t)}{rh_2(t) + h_1(t)} \quad (2.4)$$

where $h_1(t) = \frac{f(t)}{F(t)}$ and $h_2(t) = \frac{g(t)}{G(t)}$ are the failure rates of X and Y respectively. In the same manner by truncating the distribution above t , one can look at the measure in $(0, t]$. In this case

$$I_t^* = \frac{1}{r} \int_0^t \frac{f(x)}{F(t)} \left[1 - \left(\frac{g(x)}{G(t)} \right)^r \right] dx, \quad 0 < x \leq t.$$

As before we have the identity

$$1 - rI_r^*(t) = \frac{\lambda_1(t)\lambda_2^r(t) + rI_r^{*'}(t)}{r\lambda_2(t) + \lambda(t)} \quad (2.5)$$

where $\lambda_1(t) = \frac{f(t)}{F(t)}$ and $\lambda_2(t) = \frac{g(t)}{G(t)}$ are the reversed failure rates of X and Y respectively. The representation (2.4) and (2.5) are useful in characterisation problems as will be seen in the next section.

3. Characterizations

In this section we prove characterisations of some distributions based on the properties of the generalised measure of inaccuracy. Recall that if the failure rate $h_1(x)$ and $h_2(x)$ satisfy the relationship $h_2(x) = \theta h_1(x)$, $\theta > 0$ or equivalently $\overline{G}(x) = \overline{F}^\theta(x)$, then G is called the proportional hazard (PH) model with respect to F . Characterisation of PH models by properties of directed divergence measure and its generalisation $D_r(F, G)$ are discussed in Ebrahimi and Kirmani (1996); Nanda and Paul (2006), while similar results in terms of the inaccuracy measure can be found in Nair and Gupta (2007).

When $G(x)$ is the PH model of $F(x)$, equation (2.5) reduces to

$$1 - rI_r(t) = \frac{\theta^r h_1^{r+1} - rI_r'(t)}{(1 + r\theta) h_1(t)} \quad (3.1)$$

Theorem 3.1 If $F(x)$ and $G(x)$ are absolutely continuous distribution functions with G as the PH model of F , then the relationship

$$1 - rI_r(t) = k(h_1(t))^r \quad (3.2)$$

holds for any $r > -1$ if and only if $F(\cdot)$ is the generalised Pareto distribution with survival function

$$\overline{F}(t) = \left(1 + \frac{at}{b}\right)^{-\left(1+\frac{1}{a}\right)}; \quad x > 0, \quad a > -1, \quad b > 0$$

and k is some positive constant that depends on r and θ .

Proof. The generalised Pareto distribution has failure rate

$$h_1(t) = \frac{a+1}{at+b}$$

and

$$1 - rI_r(t) = \frac{\theta^r (a+1)^{r+1} (b+at)^{-r}}{[(a+1)(1+r\theta) + ar]}.$$

Hence (3.2) holds with $k = (a + 1)\theta^r[(a + 1)(1 + r\theta) + ar]^{-1}$. Conversely if we assume (3.2) we have from (3.1),

$$h_1^{r+1}(t)((1 + r\theta)k - \theta^r) = rk h_1^{r-1}(t) h_1'(t)$$

leading to

$$\frac{h_1'(t)}{h_1(t)} = \frac{(1 + r\theta)k - \theta^r}{rk}.$$

Integrating

$$h_1(t) = (c_1 t + c_2)^{-1}$$

where $c_1 = (rk)^{-1}[\theta^r - (1 + r\theta)k]$ and $c_2^{-1} = h_1(0)$. Since the failure rate of the generalized Pareto model has the above form our proof is complete ■

Remark 3.1. When $\theta = 1$, we have $F = G$ or $I_r(t) = \frac{1}{r} \left[1 - \int_t^\infty \frac{f^{(r+1)}(x)}{F^{(r+1)}(t)} \right]$, the generalised residual entropy considered in Nanda and Paul (2006). Their Theorem 4.4 is a particular case of the above result. When $r \rightarrow 0$ also we have the usual residual entropy and the result in Theorem 3.1 of Asadi and Ebrahimi (2000).

Theorem 3.2 Let $F(x)$ and $G(x)$ be absolutely continuous distribution functions with finite means. If $G(x)$ is the PH model of $F(x)$, then $F(x)$ has generalised Pareto form (3.2) if and only if

$$1 - rI_r(t) = c(m(t))^{-r}$$

where c is a constant depending on r and θ and $m(t) = E(X - t | X > t)$, the mean residual life function of X .

Proof. The proof follows from the property that $h_1(t)m(t) = c =$ a constant, characterises the generalised Pareto distribution and Theorem 3.1. ■

Remark 3.2. The result in Theorem 3.2 of Asadi and Ebrahimi (2000) is a special case of Theorem 3.2 when $\theta = 1$ and $r \rightarrow 0$.

Remark 3.3. The generalized Pareto distribution tends to the exponential distribution as $a \rightarrow 0$. Other members of the family are the Pareto distribution

specified by

$$\overline{F}(x) = \beta^c (x + \beta)^{-c}; \quad x > 0, \quad c, \quad \beta > 0$$

when $a = (\beta - 1)^{-1}$ and $b = ac$ and the rescaled beta specified by

$$\overline{F}(x) = \left(1 - \frac{x}{R}\right)^c, \quad 0 \leq x \leq R, \quad c, \quad R > 0$$

where $a = -(1 + c)^{-1}$ and $b = R(1 + c)^{-1}$.

Next we present a characterisation involving proportional reverse hazard rate model. Recall that G is said to be the proportional reversed hazard rate model of F if $\lambda_2(x) = \theta_1(x)$ or equivalently $G(x) = F^\theta(x)$. Under this model (2.5) simplifies to

$$1 - rI_r^*(t) = \frac{\theta^r \lambda_1^{r+1}(t) + rI_r^{*'}(t)}{(1 + r\theta) \lambda_1(t)} \quad (3.4)$$

Theorem 3.3 Let X and Y admit absolutely continuous distribution functions and that G be the proportional reversed hazard model of F . Then the relationship

$$1 - I_r^*(t) = k_1 (\lambda_1(t))^r \quad (3.5)$$

where k_1 is a constant depending on θ and r is satisfied if and only if X has power distribution with

$$F(t) = \left(\frac{t}{b}\right)^c, \quad 0 \leq t \leq b \quad (3.6)$$

Proof. Since, for the power distribution

$$\lambda(t) = \frac{c}{t}$$

and

$$1 - rI_r^*(t) = \frac{c\theta^r \left(\frac{c}{t}\right)^r}{c + (c\theta - 1)r}$$

(3.5) is true with $k = [c + (c\theta - 1)r]^{-1} c\theta^r$. Conversely assuming (3.5), we have from (3.4)

$$k\lambda_1^r(t) = [\theta^r \lambda_1^{r+1}(t) + kr\lambda_1^{r-1}(t) \lambda_1'(t)] [(1 + r\theta) \lambda_1(t)]^{-1}$$

which simplifies to

$$\lambda_1^{r-1}(t) [(\theta^r - (1 + r\theta)k_1)\lambda_1^2(t) - kr\lambda_1'(t)] = 0$$

or

$$\frac{\lambda_1'(t)}{\lambda_1^2(t)} = \frac{\theta^r - (1 + r\theta)k_1}{kr}.$$

Integrating

$$\theta_1(t) = \frac{1}{pt + q} \quad \text{where } p = \frac{(1 + r\theta)k - \theta^r}{kr}$$

and q is the constant of integration. Thus by the definition of $\theta_1(t)$

$$F(t) = (pt + q)f(t).$$

Since $F(0) = 0$, as $t \rightarrow 0$, $q = 0$. Thus $\lambda_1(t) = \frac{1}{pt}$. Since the reversed hazard rate characterises the distribution, $\lambda_1(t)$ corresponds to a power distribution with parameter p^{-1} . This completes the proof. ■

Remark 3.4. When $\theta = 1$, we have $F = G$ and

$$I_r^*(t) = \int_0^t \frac{f(x)}{F(t)} \left(1 - \frac{f^r(x)}{F^r(t)}\right) dx = H_F^*(t)$$

which is the truncated version of the generalised Shannon's entropy that measures the uncertainty in the spent (past) life of a unit (see Di Crescenzo and Longobardi, 2002 for details). In this case the characterisation of the power distribution in terms of $H_F^*(t)$ is by the property $1 - rH_F^*(t) = c_1\lambda_1^r(t)$ for all $t > 0$ and $R > -1$.

Theorem 3.4 Under conditions of Theorem 3.3,

$$1 - I_r^*(t) = k_2(\mu(t))^{-r} \quad (3.7)$$

where k_2 is a constant depending on θ and r and

$$\mu(t) = E(t - X | x \leq t) = [F(t)]^{-1} \int_0^t F(x) dx \quad (3.8)$$

is the reversed mean residual life function of X if and only if X has the power distribution (3.6).

Proof. The relationship between reversed mean residual life and reversed hazard rate is (Finkelstein, 2002)

$$\lambda(t) = \frac{1 - \mu'(t)}{\mu(t)}.$$

Then $\lambda(t) \mu(t) = c_3$, a constant if and only if $1 - \mu'(t) = c_3$. This gives

$$\mu(t) = (1 - c_3)t + k_3$$

where $k_3 = 0$ as $t \rightarrow 0$. Using the definition (3.8) we get

$$\frac{F(t)}{\int_0^t F(x) dx} = \frac{1}{(1 - c_3)t}$$

or

$$\log \int_0^t F(x) dx = (1 - c_3)^{-1} \log t.$$

This gives

$$F(t) = kt^{\frac{1-c_3}{c_3}},$$

the power distribution function. Conversely if $F(t)$ has the above form $\lambda(t) \mu(t) = c_3$. Thus from Theorem 3.3, X has power distribution if and only if (3.7) holds.

In the case of the PH models, the explicit relationship between F and G become handy to characterise the distribution of F in terms of $I_r(t)$. However in situations where such relations are absent we can characterise F and G only mutually, in terms of $I_r(t)$. We illustrate the case for the exponential, which can be adopted for other distributions as well. ■

Theorem 3.5 Let $F(t)$ and $G(t)$ be absolutely continuous. If $I_r(t) = c$, is a positive constant with $cr < 1$, then $F(t)$ is exponential $\left((1 - rc)^{1/r} (1 - r)^{(r-1)/r}\right)$ if and only if $G(t)$ is exponential $\left(\frac{1 - rc}{1 - r}\right)^{1/r}$.

Proof. Assume that $F(t)$ is exponential as stated. Then from (2.4)

$$1 - rc = \frac{\lambda h_2^r(t)}{\lambda + r h_2(t)}, \quad \lambda = (1 - rc)^{\frac{1}{r}} (1 - r)^{\frac{r-1}{r}}.$$

Differentiating

$$h_2'(t) [\lambda r h_2^{r-1}(t) - r(1 - rc)] = 0.$$

The solutions are $h_2(t) = \left(\frac{1 - rc}{1 - r}\right)^{\frac{1}{r}}$ and $h_2'(t) = 0$ or $h_2(t) =$ a constant.

If the second value is different from the first, it does not provide $I_r(t) = c$ and hence inadmissible. Thus G is exponential with parameter $\left(\frac{1 - rc}{1 - r}\right)^{\frac{1}{r}}$. When G is exponential, proceeding as before the resulting differential equation has only one solution

$$h_1(t) = \frac{r\beta(1 - rc)}{rc + \beta^r - 1}, \quad \beta = \left(\frac{1 - rc}{1 - r}\right)^{\frac{1}{r}}.$$

This being a constant, F is exponential with parameter $(1 - rc)^{1/r} (1 - r)^{(r-1)/r}$, on simplifying the expression for $h_1(t)$ using the value of β . ■

In conclusion, we have presented characterisation of distributions using the generalised inaccuracy measure by involving reliability functions. Though these functions originate from reliability analysis, they are used and interpreted differently in other areas like actuarial science, demography, bio-sciences, survival analysis and extreme value theory. Hence our results are applicable in a more general context in studying the uncertainty involved in modelling problems.

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